pdac_demo

Section 1B: sample and variant−level genotyping quality control

Headline Counts

Metric	Value
Raw samples	1,461
Final samples	1,215
Sample retention	83.2%
Raw variants	430,000
Final variants	414,662
Variant retention	96.4%

Report Index

This report is produced from the output files generated by Steps 01−08. It combines the count table, filtering rationale, and QC figures into one portable PDF that can be reviewed before moving to PCA and association testing.

1. QC Pipeline Count Table

Step 09

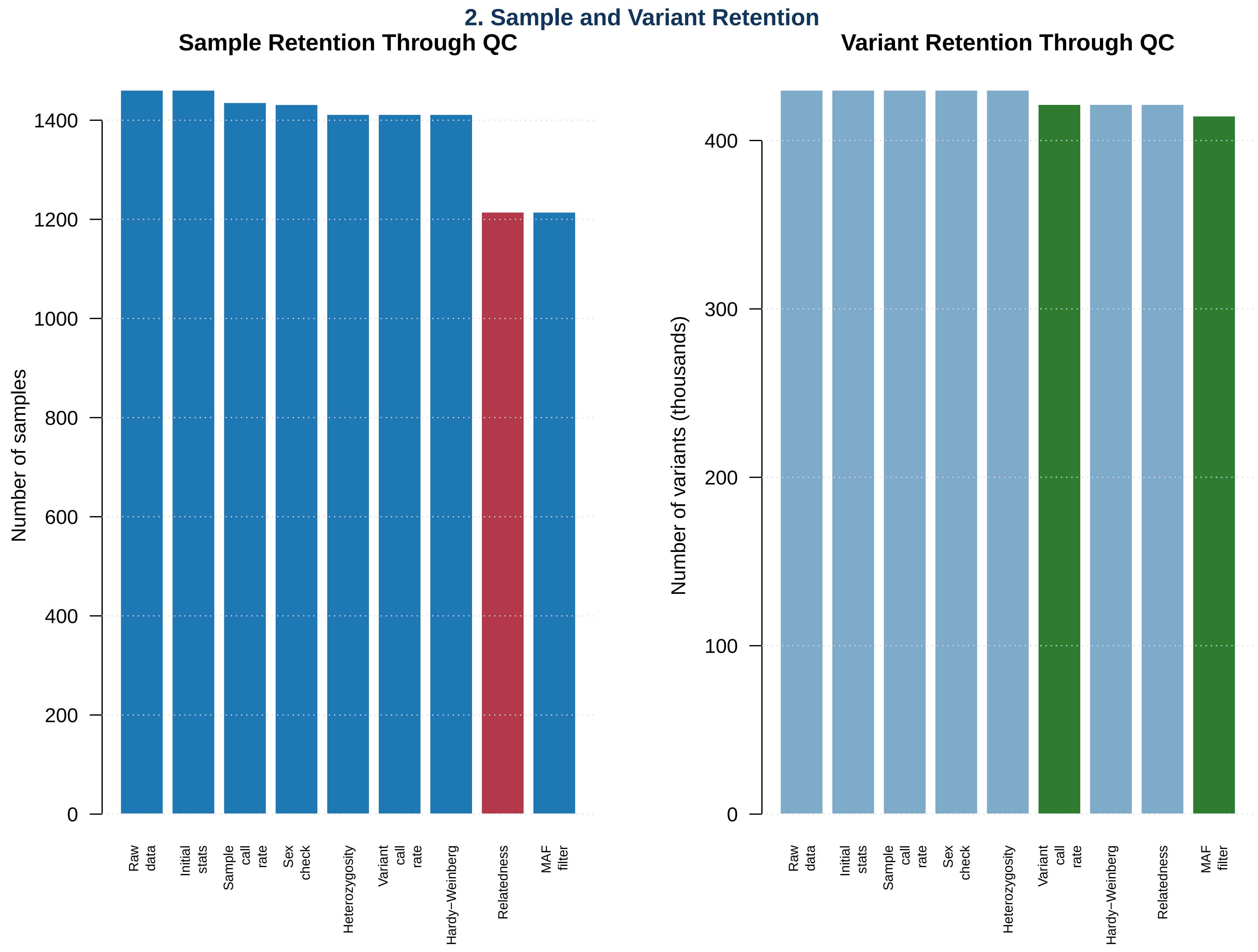
Filtering trajectory

Step	Samples	Variants	Samples lost	Variants lost
00. Raw data	1,461	430,000	0	0
01. Initial stats	1,461	430,000	0	0
02. Sample call rate	1,436	430,000	25	0
03. Sex check	1,432	430,000	4	0
04. Heterozygosity	1,412	430,000	20	0
05. Variant call rate	1,412	421,520	0	8,480
06. Hardy−Weinberg	1,412	421,520	0	0
07. Relatedness	1,215	421,520	197	0
08. MAF filter	1,215	414,662	0	6,858

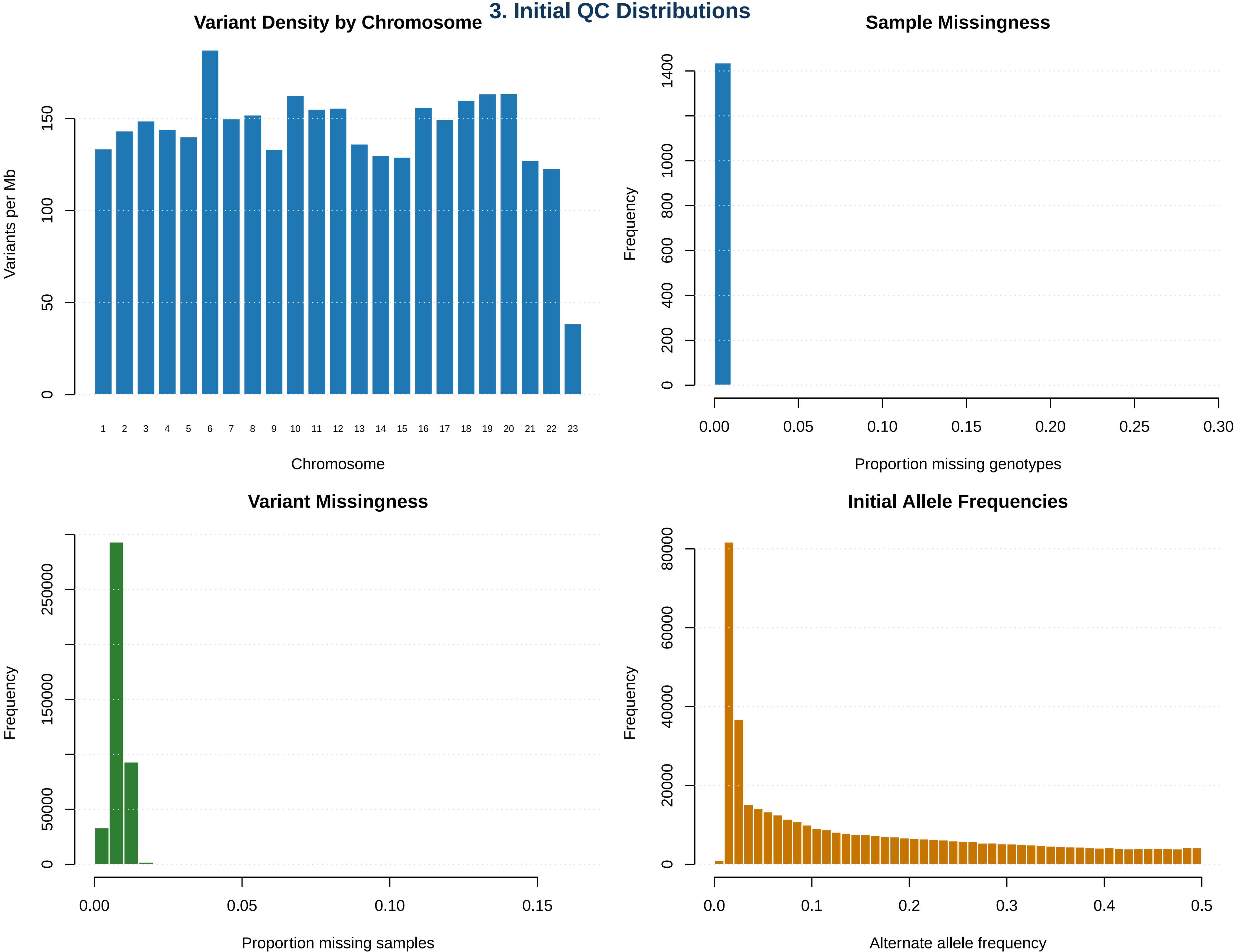
Total sample loss: 246 (16.8%)

Total variant loss: 15,338 (3.6%)

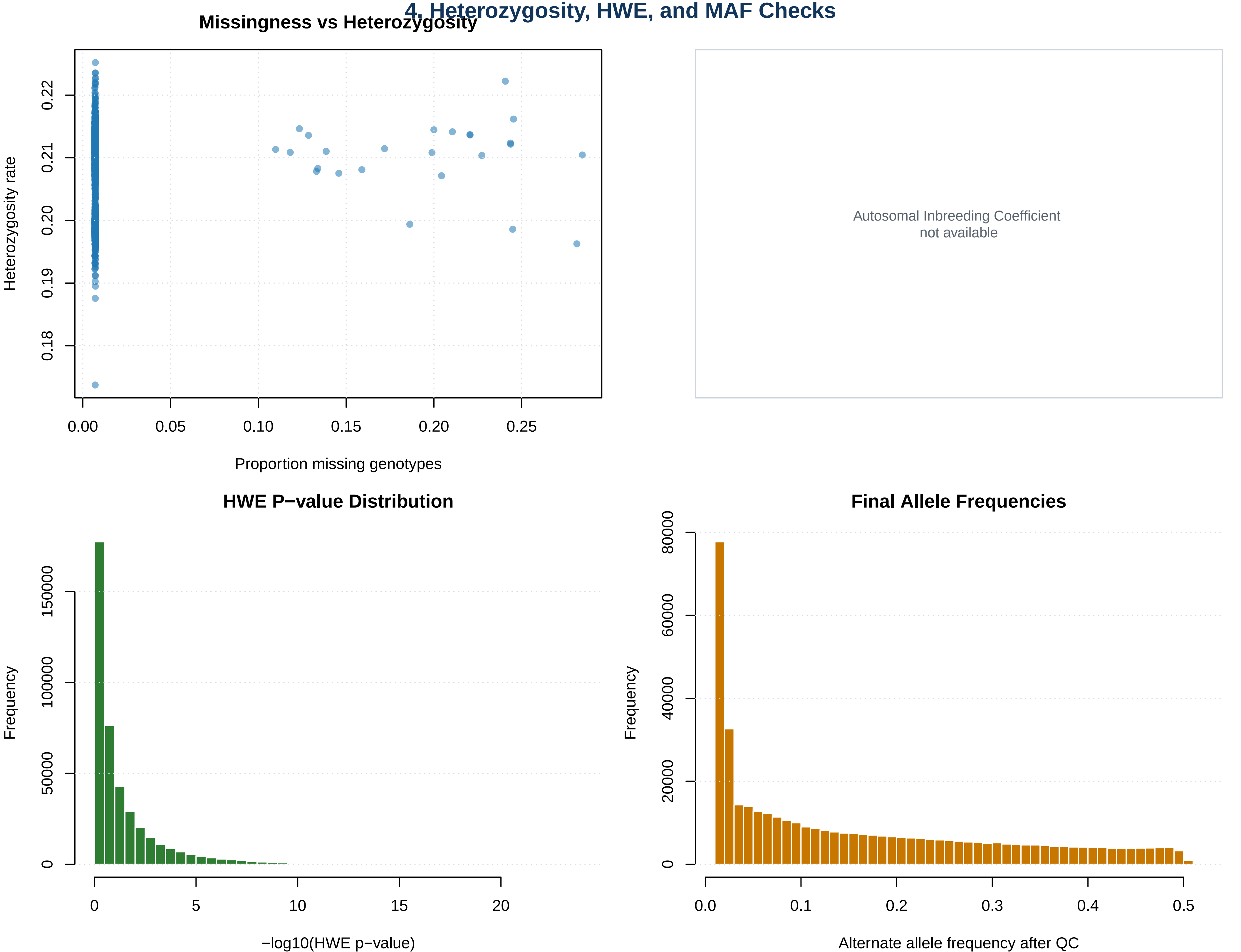
Interpretation: sample−level filters remove low−quality or non−independent individuals; variant−level filters remove unreliable markers before downstream population structure and association analyses.


2. Sample and Variant Retention
Sample Retention Through QC
Number of samples
0
200
400
600
800
1000
1200
1400
Raw data
Initial stats
Sample call rate
Sex check
Heterozygosity
Variant call rate
Hardy−Weinberg
Relatedness
MAF filter
Variant Retention Through QC
Number of variants (thousands)
0
100
200
300
400
Raw data
Initial stats
Sample call rate
Sex check
Heterozygosity
Variant call rate
Hardy−Weinberg
Relatedness
MAF filter

3. Initial QC Distributions



4. Heterozygosity, HWE, and MAF Checks



5. Relatedness Pruning Diagnostics

KING + PI_HAT

Headline diagnostics

Metric	Value
samples_before_relatedn...	1412
king_pruning_pairs_gt_0...	280
unique_samples_in_king_...	443
relatedness_components	163
largest_component_samples	6
largest_component_pairs	5
plink2_king_cutoff_remo...	166
phenotype_aware_removed...	197
phenotype_aware_removed...	14
controls_removed	167

KING related−pair summary

Relationship	Pairs	Unique samples
Duplicate or twin	0	0
First degree	280	443
Second degree	4	8
Third degree	10	20
No third degree or clos...		951
Total samples before re...		1412

Largest relatedness components

component id	n samples	n pairs	n controls	n cases	n unknown	max kinship	mean kinship
83	6	5	6	0	0	0.251375	0.247557
122	4	3	2	2	0	0.255724	0.249127
1	3	2	2	1	0	0.249054	0.249013
10	3	2	0	3	0	0.250613	0.25024
100	3	2	3	0	0	0.251349	0.250368
101	3	2	1	2	0	0.250072	0.249193

control case unknown

KING vs PI_HAT agreement

Metric	Value
total_pairs_in_either_r...	55297
pairs_reported_by_both	294
same_category	290
different_category_chec...	4
reported_by_king_only	0
reported_by_pihat_only	55003
pruning_pairs	280
pruning_pairs_with_pihat	280

Showing first 8 rows.

6. Methods Notes and Interpretation

Review before downstream analysis

QC thresholds

Level	Check	Threshold	Rationale
Sample	Call rate	−−mind 0.02	Remove low−quality samples
Sample	Sex check	X chromosome F statistic	Detect swaps or metadat...
Sample	Heterozygosity	F +/− 3 SD	Detect contamination/ou...
Sample	Relatedness	KING kinship > 0.1875	Preserve independence; ...
Variant	Call rate	−−geno 0.05	Remove poorly genotyped...
Variant	Hardy−Weinberg	−−hwe 1e−6 in controls	Remove likely genotypin...
Variant	Minor allele frequency	−−maf 0.01	Keep low−frequency sign...

Review notes

− Rare cancer setting: cases are precious, so phenotype−aware relatedness pruning is used.

− KING is used for final pruning because it is robust for relationship inference in GWAS QC; PI_HAT is reported for review and teaching.

− A large relatedness loss should trigger a review of recruitment design, family structure, and downstream relatedness−aware association options.

− The final dataset from Step 08 is the starting point for ancestry analysis, PCA, and association testing.

7. Plain−Text Summary Appendix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PDAC GWAS Genotyping Quality Control Summary
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Dataset: pdac_demo
Analysis date: Fri Jul 31 12:46:01 UTC 2026
Genome build: GRCh38

QC Pipeline Summary
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Step                          Samples    Variants   Samples lost   Variants lost
00. Raw data                  1461      430000     0              0
01. Initial stats             1461      430000     0              0
02. Sample call rate          1436      430000     25             0
03. Sex check                 1432      430000     4              0
04. Heterozygosity            1412      430000     20              0
05. Variant call rate         1412      421520     0              8480
06. Hardy−Weinberg            1412      421520     0              0
07. Relatedness (KING)        1215      421520     197             0
08. MAF filter (>=1%)         1215      414662     0              6858

Final dataset: 1215 samples and 414662 variants
Total sample loss: 246 (16.8%)
Total variant loss: 15338 (3.6%)
Sample retention: 83.2%
Variant retention: 96.4%

QC Thresholds Applied
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Sample−level:
− Sample call rate: −−mind 0.02
− Sex check: X chromosome F−statistic discordance
− Heterozygosity: autosomal F +/− 3 SD
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Appendix truncated in PDF; see the full text summary file for complete notes.